

Tithika (████████)

Post-v1 Feature Roadmap

Requirements currently not planned for v1 · May 2026 · v0.1

Introduction

This document captures all requirements and feature ideas that are explicitly **out of scope for Tithika v1**, organized into a prioritized roadmap for future versions. Items are grouped by theme and assigned a suggested release phase. Priority is indicative — actual prioritization should be driven by user feedback after v1 launch.

v1 delivers: tithi, nakshatra, special tithi indicators, the full festival calendar, localization in three scripts, offline-first operation, sunrise/sunset times, and programmatic moon rendering on both iOS and Android.

Phase Legend

Phase	Timing (indicative)	Description
v2	3-6 months post-launch	High-value enhancements that complete the core experience
v3	6-12 months post-launch	Power-user features and regional expansion
Future	12+ months / unscheduled	Aspirational or complex features requiring more research

Phase v2 — High-Value Enhancements

Notifications & Reminders

Allow users to opt in to push notifications for upcoming observances. This is among the most requested features for calendar-style apps — the ability to be reminded in advance of Ekadashi, Purnima, Amavasya, and major festivals.

Feature	Detail	Phase
Ekadashi reminder	Configurable lead time (e.g., 1 day before)	v2
Purnima reminder	Configurable lead time	v2
Amavasya reminder	Configurable lead time	v2
Festival reminders	User selects which festivals to follow	v2
Pradosh reminder	Trayodashi both pakshas	v2

Home-Screen Widgets

Native widgets (iOS WidgetKit and Android App Widgets) showing today's tithi, paksha, and sunrise time at a glance — high-visibility feature that drives daily engagement.

Widget	Content	Phase
Small widget	Tithi name + number + paksha color	v2
Medium widget	Tithi + nakshatra + sunrise/sunset	v2
iOS WidgetKit	Lock-screen and home-screen support	v2
Android widget	Home-screen support	v2

Moonrise & Moonset Times

Sunrise and sunset are shown in v1. Moonrise and moonset were explicitly deferred but are a natural complement. The Swiss Ephemeris already supports this via `swe_rise_trans`, so the implementation cost is low.

- Display moonrise and moonset times on the main screen alongside sunrise/sunset.
- Low implementation cost — same Swiss Ephemeris call already used for sunrise.

Yoga & Karana (Completing the Panchanga)

Yoga and Karana are the remaining two of the five Panchanga elements (Tithi, Nakshatra, Yoga, Karana, and Vara). Their inclusion would offer a complete Panchanga view.

Element	Calculation	Phase
Yoga	$(\text{moonLong} + \text{sunLong}) / (360/27)$ — one of 27 yogas	v2

Karana	Half-tithi unit, 11 types (7 movable + 4 fixed)	v2
Vara (weekday)	Already implicit — surface explicitly with Sanskrit name	v2

Phase v3 — Power-User Features & Regional Expansion

Inauspicious Time Periods

Rahu Kaal, Yamaganda, Gulika, and Abhijit muhurta are commonly referenced in Hindu religious practice. These are time-of-day divisions that depend on weekday (Vara) and are straightforward to compute once sunrise is known.

Feature	Basis	Phase
Rahu Kaal	1.5-hour window, varies by day of week	v3
Yamaganda	1.5-hour window, complementary to Rahu Kaal	v3
Gulika Kaal	1.5-hour window	v3
Abhijit Muhurta	Auspicious window around solar noon	v3

Regional Calendar Variants

Different regions of India use different conventions for month names, festival dates, and calendar systems. Expanding regional support is essential to serve a broader user base.

Region / System	Key Difference	Phase
Amanta month system	Month ends on Amavasya — toggle already planned for v1 Settings	v1 toggle
Tamil calendar	Different month names; Onam detected by nakshatra (Thiruvonam)	v3
Bengali calendar	Shakabda era, different month names	v3
Malayalam / Kollavarsham	Different epoch and month system	v3
Marathi calendar	Closely follows standard but with regional festival differences	v3
Gujarati calendar	Vikram Samvat, different New Year (Bestu Varas)	v3

Multi-Region Festival Date Variations

Several major festivals are observed on different dates in different regions. v3 should allow users to select their regional convention.

- Add a "Region" setting (e.g., North India, South India, Maharashtra, Bengal).
- Festival dates adjust automatically based on the region selection.

Additional Languages

Expanding the language offering to cover major Indian languages with their own scripts:

- Tamil (Tamil script)
- Bengali (Bengali script)
- Malayalam (Malayalam script)
- Marathi (Devanagari — distinct vocabulary from Hindi)
- Gujarati (Gujarati script)

Vakya Calculation Method

Vakya (or Vakya Siddhanta) is a traditional algorithmic method used in South Indian Panchangas, as opposed to Drik (direct astronomical observation). Offering both methods side-by-side would serve traditional practitioners.

- Add a "Calculation method" toggle in Settings: Drik (default) / Vakya.
- Clearly label all outputs with the active method.

Future Phase — Aspirational Features

Muhurta (Auspicious Time Calculations)

Full muhurta calculation combines tithi, nakshatra, yoga, karana, and lagna (ascendant) to identify auspicious windows for specific activities (marriage, travel, business). This is computationally intensive and may require a backend.

- Marriage muhurta, Griha Pravesh muhurta, Namkaran muhurta, and more.
- Requires lagna computation (rising sign), needing time-stepped sidereal calculation.

Photo-Realistic Moon Imagery

v1 uses a programmatic moon render based on illumination percentage. A future version could use high-resolution pre-rendered moon images or real NASA photography for each phase.

- Pre-render ~30 moon images (one per tithi) at high resolution.
- Or composite real lunar surface photography with programmatic phase masking.

Monetization

v1 is explicitly ad-free with no in-app purchases. Monetization is deferred until the product has established a user base. Possible future models:

- Optional "Supporter" in-app purchase (no feature gating).
- Premium tier for muhurta, regional calendars, or advanced notifications.
- No advertising — this is a deliberate product philosophy decision.

Full Panchanga View

A dedicated screen showing all five Panchanga elements (Tithi, Nakshatra, Yoga, Karana, Vara) alongside Rahu Kaal, Yamaganda, Gulika, and Abhijit — mirroring the format of a traditional printed Panchanga.

Pending Design Decisions (from v1 Open Items)

These items are unresolved and must be addressed before or during the design phase:

Open Item	Current Draft / Notes	Target Resolution
Color palette refinement	Shukla: #d9a441 · Krishna: #4a5fc1 · Accent: #c8421d	Design phase
Typography (Devanagari)	Evaluate Noto Sans Devanagari for full glyph coverage	Design phase
Moon image animation	Static vs. animated phase transition — impacts battery and complexity	Design phase

Roadmap Summary

Feature	Phase	Effort	Dependencies
Notifications & reminders	v2	Medium	OS push notification APIs
Home-screen widgets	v2	Medium	iOS WidgetKit, Android AppWidget
Moonrise & moonset	v2	Low	Swiss Ephemeris (already integrated)
Yoga & Karana	v2	Low-Medium	Swiss Ephemeris
Rahu Kaal / Yamaganda / Gulika	v3	Low	Sunrise (already in v1)
Abhijit Muhurta	v3	Low	Sunrise (already in v1)
Vakya calculation method	v3	High	Vakya algorithm implementation
Tamil calendar	v3	High	Onam detection + Tamil localization
Bengali / Malayalam / Marathi / Gujarati	v3	High	Regional research + localization
Multi-region festival variations	v3	Medium	Festival data expansion
Additional languages	v3	Medium	Translation + script rendering
Full Panchanga view	Future	Medium	Yoga + Karana from v2
Muhurta calculations	Future	Very High	Lagna computation + extensive testing
Photo-realistic moon imagery	Future	Medium	Art direction / photography
Monetization	Future	Low-Medium	Post-launch user base establishment